

News Topic Classification from Headlines: A Multi-Model Comparative Study

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Abstract—This study explores the classification of news using a data set of 117,710 headlines in four categories: Business, Science and Technology, World News, and Sports. Initial analysis revealed significant noise, such as HTML tags and stop words, so three text versions were created to test varying levels of cleaning and stemming. Vectorization methods, including TF-IDF and Skipgram, were applied to convert text into numerical features. Performance was evaluated across several models, including Logistic Regression, Deep Neural Networks (DNN), and recurrent neural networks such as SimpleRNN, GRU, and LSTM. The results show that the models generally performed best on either the raw text or the extreme stemming datasets. The DNN achieved the highest overall accuracy of 0.9169 when combined with TF-IDF and extreme stemming, while logistic regression reached the accuracy of 0.9140 using raw text. These findings suggest that while extensive preprocessing can benefit certain neural architectures, raw text often retains enough context for high-quality news categorization.

Index Terms—Artificial neural networks; Feature extraction; Information filtering; Logistic regression; Machine learning; Natural language processing; News classification; Recurrent neural networks; Stemming; Text processing; TF-IDF.

I. INTRODUCTION

This project focuses on the automated classification of news headlines into four distinct categories which are Business, Science and Technology, World News, and Sports. The goal of this task is to make it easier to find information and help organize large news collections. The dataset used for this research contains 117,710 entries. The labels are fairly distributed across the four topics, led by Business with 29,550 entries, followed by Science and Technology with 26,679, World News with 24,747, and Sports with 14,751.

Looking closely at the data showed that the raw headlines often contained extra noise like HTML tags and common words that do not help with prediction. To fix this, the project compared three different ways of cleaning the text. These included using the raw text with no changes, an extreme version that used Porter Stemming to trim words down, and an optimum version that turned symbols into descriptive words like MONEY or PERCENT. This text was then turned into

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numbers using techniques like TF-IDF and Skipgram word embeddings.

The study tested several different computer models including Logistic Regression, Deep Neural Networks, and various types of Recurrent Neural Networks like GRU and LSTM. Most of these models showed high accuracy. The results showed that the best cleaning method depended on the specific model being used. For example, the Deep Neural Network performed best at 0.9169 accuracy when using the extreme stemming method. However, Logistic Regression performed very well on raw text with an accuracy of 0.9140. Overall, the results suggest that while special cleaning helps some models, the original headlines usually have enough information for the computer to categorize them correctly.

II. METHODOLOGY

A. Dataset and Exploratory Data Analysis

The dataset consists of two CSV files divided into training and test sets. Each file contains two columns named News Headline and News Topic. The headlines include HTML tags from the web scraping process.

Combining both files resulted in 117,710 total rows. Removing 21,983 duplicate entries reduced the count to 95,727 unique news samples. Table I summarises the class distribution.

TABLE I
CLASS DISTRIBUTION IN THE COMBINED DATASET

News Topic	Count	Share (%)
World News	46,730	39.7
Business	29,550	25.1
Science and Technology	26,679	22.7
Sports	14,751	12.5

The data distribution shows a split between categories. World News accounts for 39.7 percent while Sports accounts for 12.5 percent. This imbalance led to the use of balanced class weights in the Logistic Regression model. The macro averaged F1 score serves as the main measurement for performance. Initial word count analysis on the raw text shows an average of 47 words per headline. This number is high because

Distribution of News Topics

A bar chart titled 'Distribution of News Topics' showing the number of articles for four categories. The y-axis is labeled 'Number of Articles' and ranges from 0 to 50,000 in increments of 10,000. The x-axis is labeled 'News Topic'. The bars are colored blue for World News, light blue for Business, orange for Sci. & Tech., and light orange for Sports. The exact values are displayed above each bar.

News Topic	Number of Articles
World News	46,730
Business	29,550
Sci. & Tech.	26,679
Sports	14,751

2-gram	Frequency (approx.)
br br	17,500
html body	14,800
news headlines	14,800
body news	11,800
headlines br	11,500
in the	3,500
of the	3,200
body br	2,800
br news	2,800
br html	2,800

[illegible]

Figure 1 displays the total number of articles for each news category. Figure 2 lists the ten most common pairs of words found in the sports section. Figure 3 provides a WordCloud to visualize frequently used terms in sports. Figure 4 illustrates a Co-Occurrence Matrix to show how the ten most frequent sports words appear together in the text.

Word Co-occurrence: Sports (Top 10 words)										
and	0	7784	24852	7756	9349	6397	7848	5296	17692	7000
body	7784	0	47486	14845	17835	11818	15015	10115	31500	13748
br	24852	47486	0	47345	59265	37726	47891	32253	100655	43892
headlines	7756	14845	47345	0	17782	11791	14971	10105	31431	13697
html	9349	17835	59265	17782	0	14156	17988	12087	37881	16521
in	6397	11818	37726	11791	14156	0	11910	8493	28564	10583
news	7848	15015	47891	14971	17988	11910	0	10224	31840	13843
of	5296	10115	32253	10105	12087	8493	10224	0	26261	8712
the	17692	31500	100655	31431	37881	28564	31840	26261	0	30467
to	7000	13748	43892	13697	16521	10583	13843	8712	30467	0
	and	body	br	headlines	html	in	news	of	the	to

B. Preprocessing

1) No Preprocessing: The raw headlines still containing HTML code are used as the baseline. This text goes directly into the TF-IDF and Skip-gram vectorizers.

3) Optimum Preprocessing: This version adds special labels to the text. Specific values change into general words like MONEY, PERCENT, or UNIT. Instead of stemming, WordNet lemmatization [9] is used to keep words in a readable format. This method works best for models that use dense word embeddings.

Two vectorization methods transform text into numerical data for the models.

1) TF-IDF [1]: Used for the Logistic Regression and DNN models. This approach identifies 10,000 important features, including both single words and two-word phrases. It assigns higher weights to unique words that help distinguish different topics while lowering the weight of common terms that appear frequently across the entire dataset.

2) **Skip-gram [2]:** Used for the RNN-based models. This model is trained directly on the news headlines to create 100-dimensional word vectors using a 5-word context window. To turn these individual words into a single representation for each headline, the system uses SIF [3] weighting. This method gives more importance to rare, meaningful words and less to common ones by calculating a specific weight for each word based on its frequency, ensuring the final data captures the most relevant information.

D. Model Architectures

1) Logistic Regression [4]: This model uses L2 regularization and TF-IDF features. The best value for the parameter C is chosen from 0.1, 1.0, and 10.0 based on the best macro F1 score. The model uses balanced class weights to handle the data distribution and runs for a maximum of 1,000 iterations.

2) Deep Neural Network (MLP) [5]: A Multi-Layer Perceptron undergoes extensive tuning. Four different hidden layer structures are tested alongside three learning rates and a batch size of 2,048. Early stopping is used to prevent overfitting if performance does not improve for three iterations. The final model is retrained using the combined training and validation data.

3) Recurrent Neural Networks: Six different architectures are built using TensorFlow and Keras [10] and they involve SimpleRNN, GRU [7], LSTM [6], and their bidirectional versions. Each model passes data through the recurrent layer, followed by a dense layer with 64 units, and ends with a softmax layer for classification. These models use the Adam optimizer and a batch size of 1,024. Tuning includes testing various learning rates and hidden units.

Note: The SIF embedding method combines each headline into a single vector. Because the recurrent layers only process one time step, these models function similarly to a dense layer. This explains why the Deep Neural Network often performs better than the Recurrent Neural Networks on this specific task.

Data split: The first 83,727 rows form the training and validation pool, divided by an 85/15 percent split. The remaining 12,000 rows serve as the test set. This test set is perfectly balanced with exactly 3,000 samples for each of the four news categories.

III. RESULTS

A. Overall Performance

Table II lists the results for the 24 configurations evaluated on the test set. The overall best performance was achieved by the Deep Neural Network using TF-IDF features and Extreme Stemming, which reached an accuracy of 0.9169 and a macro F1-score of 0.9170. Logistic Regression followed closely; using raw text with no preprocessing yielded an F1-score of 0.9139, while the Optimum variant resulted in an F1-score of 0.9135.

Figure 5 compares the two best models that used TF-IDF features. Figure 6 shows the performance of the various deep learning models when using Skip-gram embeddings.

B. Effect of Preprocessing Strategy

Extreme Stemming produces the highest F1 score of 0.9170 for the Deep Neural Network when using TF-IDF. Logistic Regression is also very reliable. It keeps F1 scores between 0.9122 and 0.9139 across all three text versions.

For models using Skip-gram embeddings, Extreme Stemming has mixed results. Some models like the GRU saw a small drop in performance. However, other models like the SimpleRNN actually improved. Stemming changes words into

TABLE II
COMPLETE RESULTS: ACCURACY AND MACRO F1 FOR ALL 24 MODEL-PREPROCESSING-EMBEDDING CONFIGURATIONS. **BOLD** = GLOBAL BEST F1.

Model	Embed.	Preproc.	Acc.	F1-Mac.
LR	TF-IDF	No Prep.	0.9140	0.9139
LR	TF-IDF	Ext. Stem	0.9123	0.9122
LR	TF-IDF	Optimum	0.9137	0.9135
DNN (MLP)	TF-IDF	No Prep.	0.9038	0.9043
DNN (MLP)	TF-IDF	Ext. Stem	0.9169	0.9170
DNN (MLP)	TF-IDF	Optimum	0.9131	0.9132
SimpleRNN	Skip-gram	No Prep.	0.9024	0.9028
SimpleRNN	Skip-gram	Ext. Stem	0.9084	0.9085
SimpleRNN	Skip-gram	Optimum	0.9083	0.9084
GRU	Skip-gram	No Prep.	0.9081	0.9082
GRU	Skip-gram	Ext. Stem	0.8931	0.8933
GRU	Skip-gram	Optimum	0.9038	0.9038
LSTM	Skip-gram	No Prep.	0.9097	0.9099
LSTM	Skip-gram	Ext. Stem	0.9007	0.9010
LSTM	Skip-gram	Optimum	0.9051	0.9052
Bi-SimpleRNN	Skip-gram	No Prep.	0.9131	0.9133
Bi-SimpleRNN	Skip-gram	Ext. Stem	0.9054	0.9058
Bi-SimpleRNN	Skip-gram	Optimum	0.9116	0.9117
Bi-GRU	Skip-gram	No Prep.	0.9091	0.9091
Bi-GRU	Skip-gram	Ext. Stem	0.9054	0.9058
Bi-GRU	Skip-gram	Optimum	0.9050	0.9052
Bi-LSTM	Skip-gram	No Prep.	0.9111	0.9112
Bi-LSTM	Skip-gram	Ext. Stem	0.9084	0.9083
Bi-LSTM	Skip-gram	Optimum	0.9079	0.9080

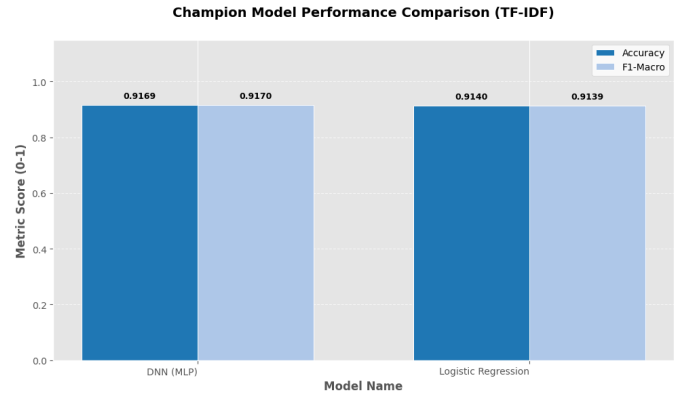


Fig. 5. Best model performance on TF-IDF features. DNN (MLP) achieves the new top F1 of 0.9170; LR follows at F1 $\approx$ 0.9139.

root forms that might not match the pre-trained vocabulary. Even so, these neural models are often strong enough to handle those changes without losing too much accuracy.

The Optimum Preprocessing method is the most consistent choice. It achieves the best or second-best F1 score in almost every test using dense embeddings. This proves that a balanced strategy is best. By removing HTML, fixing domain names, and using lemmatization [9], you create clean text that works well for both simple and complex models.

Fig. 7 displays a complete heatmap of the F1-Macro scores across all model and preprocessing combinations. This dia-

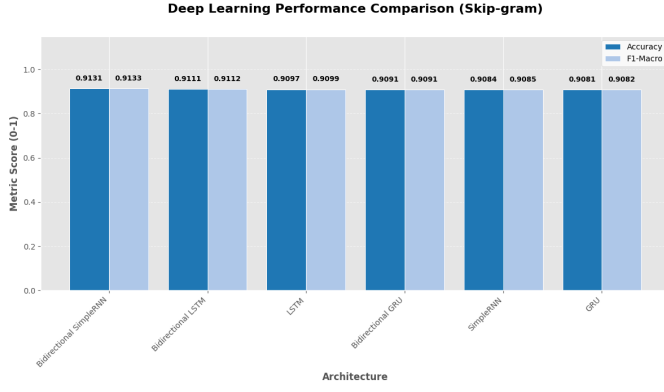


Fig. 6. Deep-learning architecture comparison on Skip-gram embeddings. Best result per model shown. Bidirectional SimpleRNN leads all RNNs in the No Preprocessing configuration.

gram provides a direct comparison of every configuration to show which pairs performed best. The visual data illustrates that specific models like the DNN achieve better results with aggressive cleaning while others like the RNN variants keep higher scores when more text context is available.

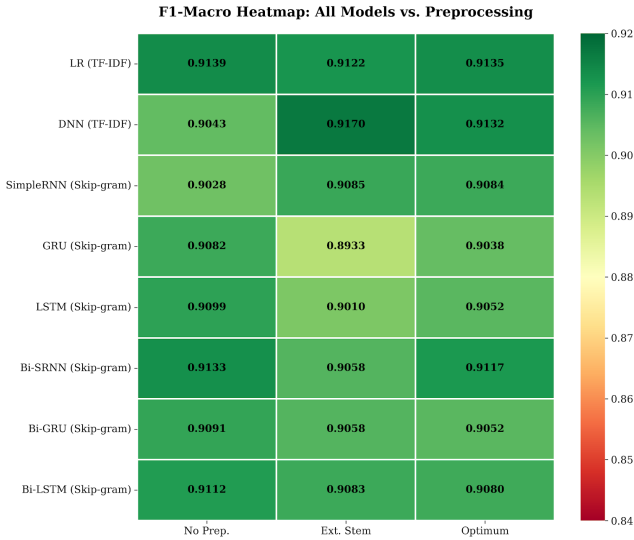


Fig. 7. F1-Macro heatmap across all 24 model preprocessing combinations. Extreme stemming (centre column) consistently underperforms for dense-embedding models.

Figure 8 illustrates the architecture of the best performing Deep Neural Network model. The diagram shows an input layer designed for 10,000 TF-IDF features followed by a sequence of three hidden layers. These hidden layers consist of 256, 128, and 64 neurons respectively and all use ReLU activation. The structure concludes with a 4-neuron output layer using Softmax activation to classify headlines into Business, Science and Tech, World News, or Sports.

C. TF-IDF vs. Dense Embeddings

The best TF-IDF result of 0.9170 outperforms the best Skip-gram result of 0.9133 by a small margin of 0.37 percentage

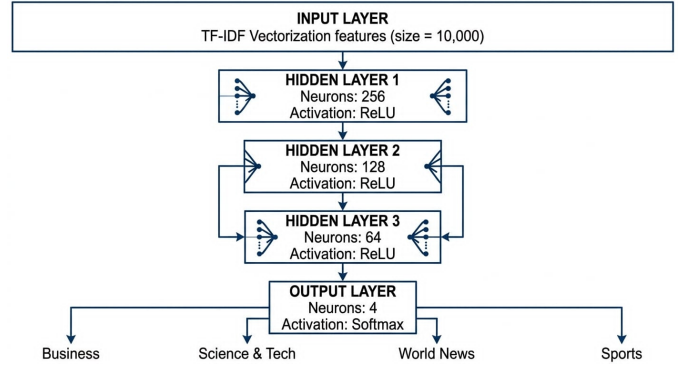


Fig. 8. Best Performing DNN model.

points (0.0037). Three factors explain why TF-IDF remains highly effective for this task. First, news headlines contain many specific named entities (such as athlete and company names) that act as strong markers and are directly encoded as high-weight features by TF-IDF [1]. Second, the SIF [3] aggregation method used for embeddings discards word order and contextual nuance. Finally, the large training set of 83,727 samples across only four classes is sufficient for TF-IDF models to learn clear decision boundaries without requiring deeper representations.

D. Deep Architecture Comparison

Among the recurrent models using raw text, the Bidirectional SimpleRNN reaches the highest F1-score of 0.9133, followed by the Bidirectional LSTM at 0.9112 and the LSTM at 0.9099. When using the Optimum preprocessing strategy, the Bidirectional SimpleRNN leads with an F1-score of 0.9117, followed closely by the SimpleRNN at 0.9084 and the Bidirectional LSTM at 0.9080. Across most categories, the bidirectional versions show marginal but consistent improvements over their unidirectional counterparts when the text is cleaned.

E. Per-Class Analysis

Table III and Figure 9 present the per-class results for the top-performing model configuration: the DNN utilizing TF-IDF features coupled with extreme stemming, which achieved a peak Macro F1-score of 0.9170. Class-specific analysis reveals that the Sports category achieves the highest overall performance due to its highly distinct vocabulary, followed closely by robust results in Science and Technology. Conversely, the primary classification errors occur between the Business and Science and Tech categories, which reflects the natural semantic overlap between economic and technology headlines. This specific architecture benefited significantly from the extreme stemming pipeline, which stripped out structural metadata, headers, and tags (leaving exclusively alphabetic characters and spaces) prior to applying the Porter Stemmer.

TABLE III
PER-CLASS PERFORMANCE REPORT FOR THE PEAK DNN MODEL USING
TF-IDF AND EXTREME STEMMING ($n = 12,000$)

Class	Precision	Recall	F1-Score	Support
Business	0.88	0.89	0.88	3,000
Science & Tech.	0.89	0.91	0.90	3,000
Sports	0.97	0.97	0.97	3,000
World News	0.94	0.89	0.91	3,000
Macro Avg.	0.92	0.92	0.92	12,000

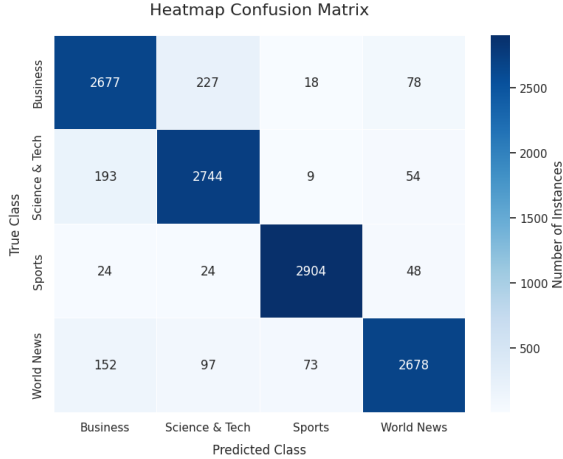


Fig. 9. Confusion matrix for the best model (DNN + TF-IDF + Extreme). Sports is the most accurate category with 96.80 percent accuracy. The biggest problem for the model is telling the difference between Business and Science and Tech, which caused 420 mistakes.

IV. CONCLUSION

This research utilized a news classification dataset containing 117,710 headlines divided into four categories: Business, Science and Technology, World News, and Sports. Detailed data analysis revealed that the raw text contained substantial structural noise, such as HTML tags and news headers. To address this, we developed three distinct preprocessing configurations: the raw original text, an extreme stemming configuration that applied a Porter Stemmer [8] to reduce words to their base forms, and an optimum preprocessing configuration that tokenized specialized symbols (e.g., converting dollar signs and percentage symbols into explicit placeholder tokens like MONEY and PERCENT).

The study evaluated the TF-IDF and Skip-gram vectorization methods alongside several deep learning and machine learning architectures to identify the most effective classification pipeline. The top-performing model configuration was a Deep Neural Network (DNN) paired with TF-IDF vectorization on the extremely stemmed text. This model achieved a peak Macro F1-score of 0.9170, outperforming the baseline Logistic Regression model. This demonstrates that with appropriate optimization, deeper neural network architectures can surpass traditional classifiers, even when processing short-text inputs like headlines.

Text preprocessing proved to have a highly variable impact

depending on the downstream architecture. While the top-performing DNN benefited significantly from the extreme stemming configuration, recurrent architectures like the GRU and LSTM exhibited performance degradation when the text was overly modified. For instance, the LSTM model experienced a notable drop in F1-score when moving from raw text to the extreme stemming pipeline. The highest performance among the recurrent models was achieved by the Bidirectional SimpleRNN using raw text paired with Skip-gram embeddings, reaching a Macro F1-score of 0.9133. This suggests that these sequence-based architectures heavily rely on the complete contextual and syntactic integrity of the original word sequences.

The study acknowledges certain limitations, including residual noise within the text corpus, the loss of word order inherent to bag-of-words vectorization techniques, and a slight class imbalance across the categories. Future work will focus on integrating advanced transformer-based representations, such as BERT, and exploring alternative embedding strategies to enhance the sequence-modeling capabilities of the recurrent architectures.

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